

Three environments, three sealed verdicts: no stable improvement on TWAP anywhere -- including where a simple rule proves the edge exists

1. Frozen replay of real order-book data

no stable edge — 4 of 14 arms clear the bar, and a diagnosed-broken learner clears it too

1-min data, 30-min deadline · DQN 97 BTC
1-min data, 30-min deadline · DQN 193 BTC
1-min data, 30-min deadline · PPO 97 BTC
1-min data, 30-min deadline · PPO 193 BTC
10-s data, 30-min deadline · DQN 97 BTC
10-s data, 30-min deadline · DQN 193 BTC
10-s data, 30-min deadline · DQN 386 BTC
10-s data, 30-min deadline · PPO 97 BTC
10-s data, 30-min deadline · PPO 193 BTC
10-s data, 30-min deadline · PPO 386 BTC
10-s data, 10-min deadline · DQN 97 BTC
10-s data, 10-min deadline · DQN 193 BTC
10-s data, 10-min deadline · PPO 97 BTC
10-s data, 10-min deadline · PPO 193 BTC



2. Reactive simulator, no injected signal

null — both sealed confirmations fail

selected config (lr 1e-3), volatile · PPO
remedial pick (bigger net), volatile · PPO



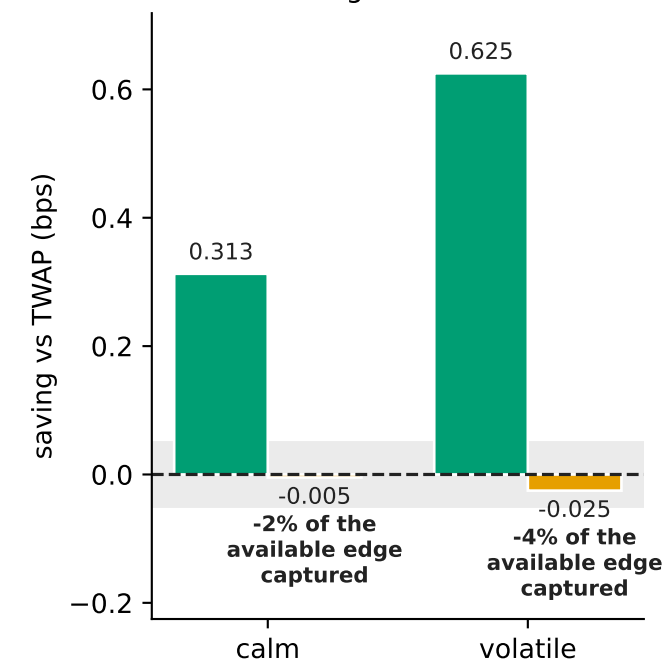
3. Reactive simulator + measured signal

null — both regimes fail

calm regime · PPO
volatile regime · PPO



Environment 3: the edge was real and the agents left all of it



● PPO arm (sealed)

■ DQN arm (sealed)

---- TWAP

±0.05 bps materiality band

■ registered rule, zero fitted parameters

■ trained agents (sealed)